

Even though GPT-4 correctly answers the simpler questions I outlined in this chapter, you can still find plenty of examples of GPT-4 failing on commonsense and theory-of-mind questions. GPT-4's lack of a world model can be seen by probing deeper with more complicated questions. But it is becoming progressively more painstaking to find these examples. In some ways, this has become a game of Whac-A-Mole; everytime a skeptic publishes examples of commonsense questions that LLMs answer incorrectly, companies like OpenAI simply use these examples as training data for the next update of their LLMs, which thereby answer such questions correctly.

Indeed, the massive size of these models, along with the astronomical quantity of data on which they are trained, in some ways obscures the underlying differences between how LLMs think and how humans think. A calculator performs arithmetic better than any human, but still lacks the same understanding of math as a human. Even if LLMs correctly answer commonsense and theory-of-mind questions, it does not necessarily mean it reasons about these questions in the same way.

As Yann Lecun said, “the weak reasoning abilities of LLMs are partially compensated by their large associative memory capacity. They are a bit like students who have learned the material by rote but haven't really built deep mental models of the underlying reality.” Indeed, these LLMs, like a supercomputer, have a gargantuan memory capacity, having read more books and articles than a single human brain could consume in a thousand lifetimes. And so what seems like commonsense reasoning is really more like pattern matching, done over an astronomically enormous corpus of text.

But still, these LLMs are an incredible step forward. What is most amazing about the success of LLMs is how much they seemingly understand about the world despite being trained on nothing but language. LLMs can correctly reason about the physical world without ever having experienced that world. Like a military cryptanalyst decoding the meaning behind encrypted secret messages, finding patterns and meanings in what was originally gibberish, these LLMs have been able to tease out aspects of a world they have never seen or heard, that they have never touched or experienced, by merely scanning the entire corpus of our uniquely human code for transferring thoughts.

It is possible, perhaps inevitable, that continuing to scale up these language models by providing them with more data will make them even better at answering commonsense and theory-of-mind questions.* But without incorporating an inner model of the external world or a model of other minds—without the breakthroughs of simulating and mentalizing—these LLMs will fail to capture something essential about human intelligence. And the more rapid the adoption of LLMs—the more decisions we offload to them—the more important these subtle differences will become.

In the human brain, language is the *window* to our inner simulation. Language is the interface to our mental world. And language is built on the foundation of our ability to model and reason about the minds of others—to infer what they mean and figure out exactly which words will produce the desired simulation in their mind. I think most would agree that the humanlike artificial intelligences we will one day create will not be LLMs; language models will be merely a window to something richer that lies beneath.